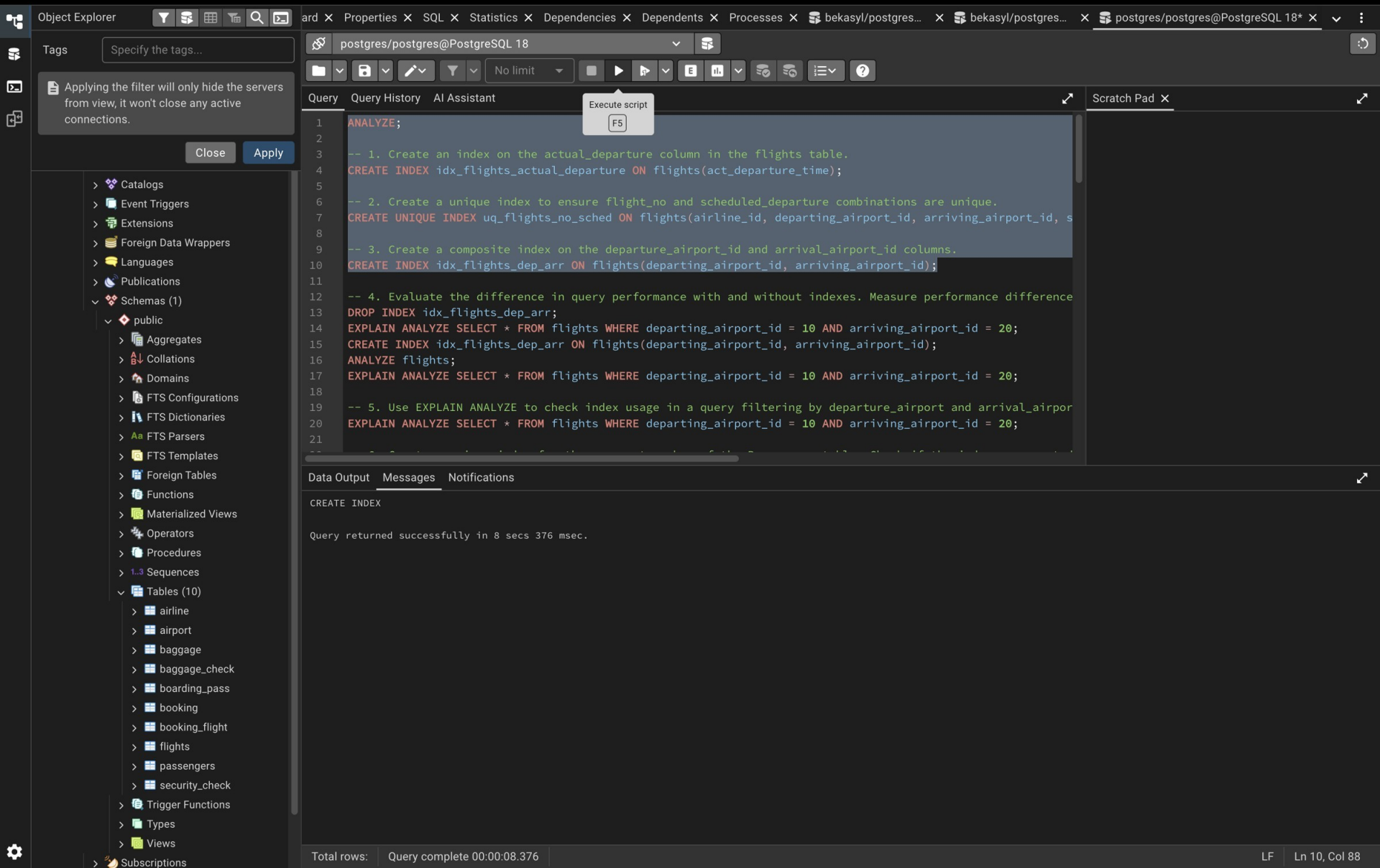
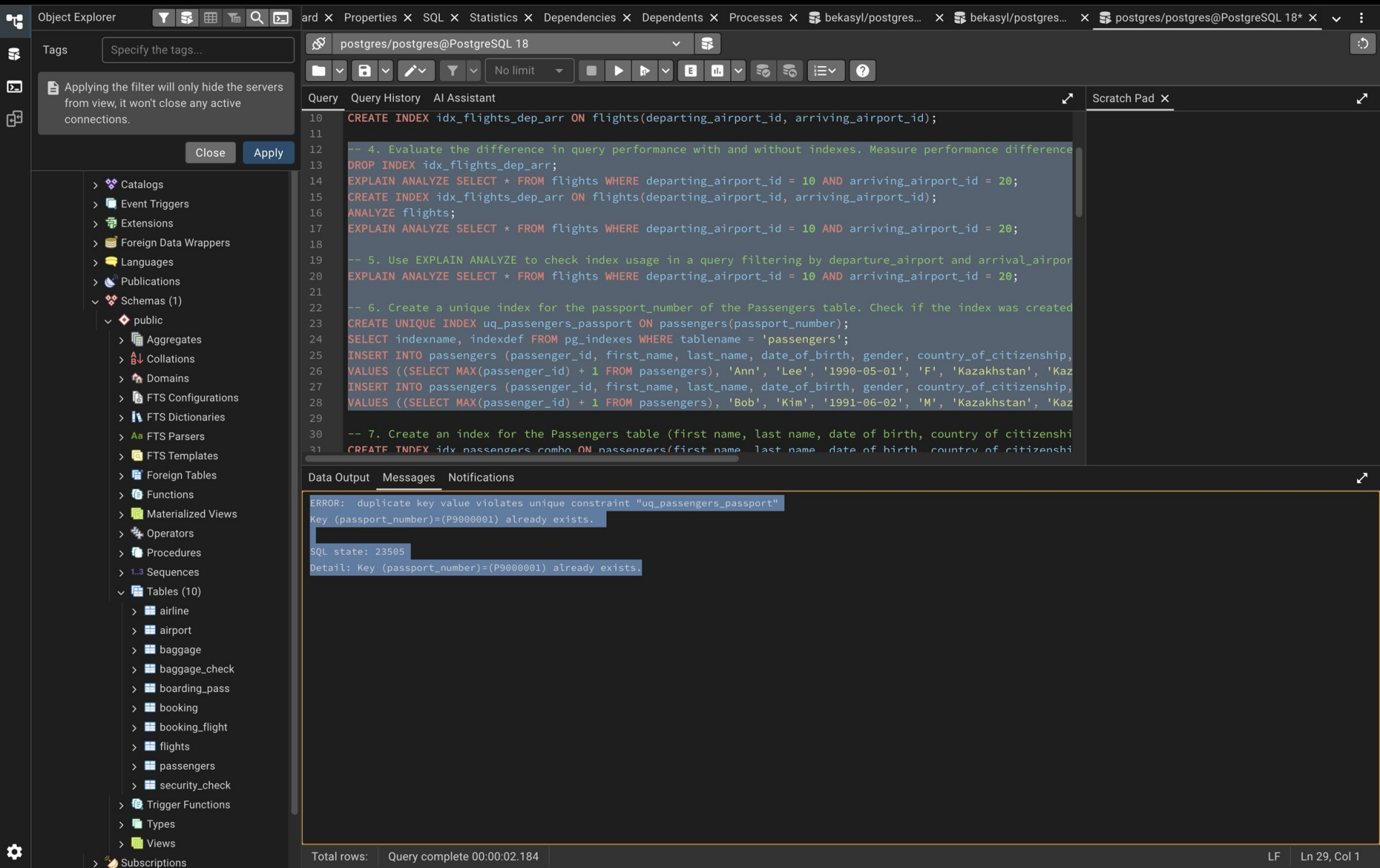
Laboratory work 7

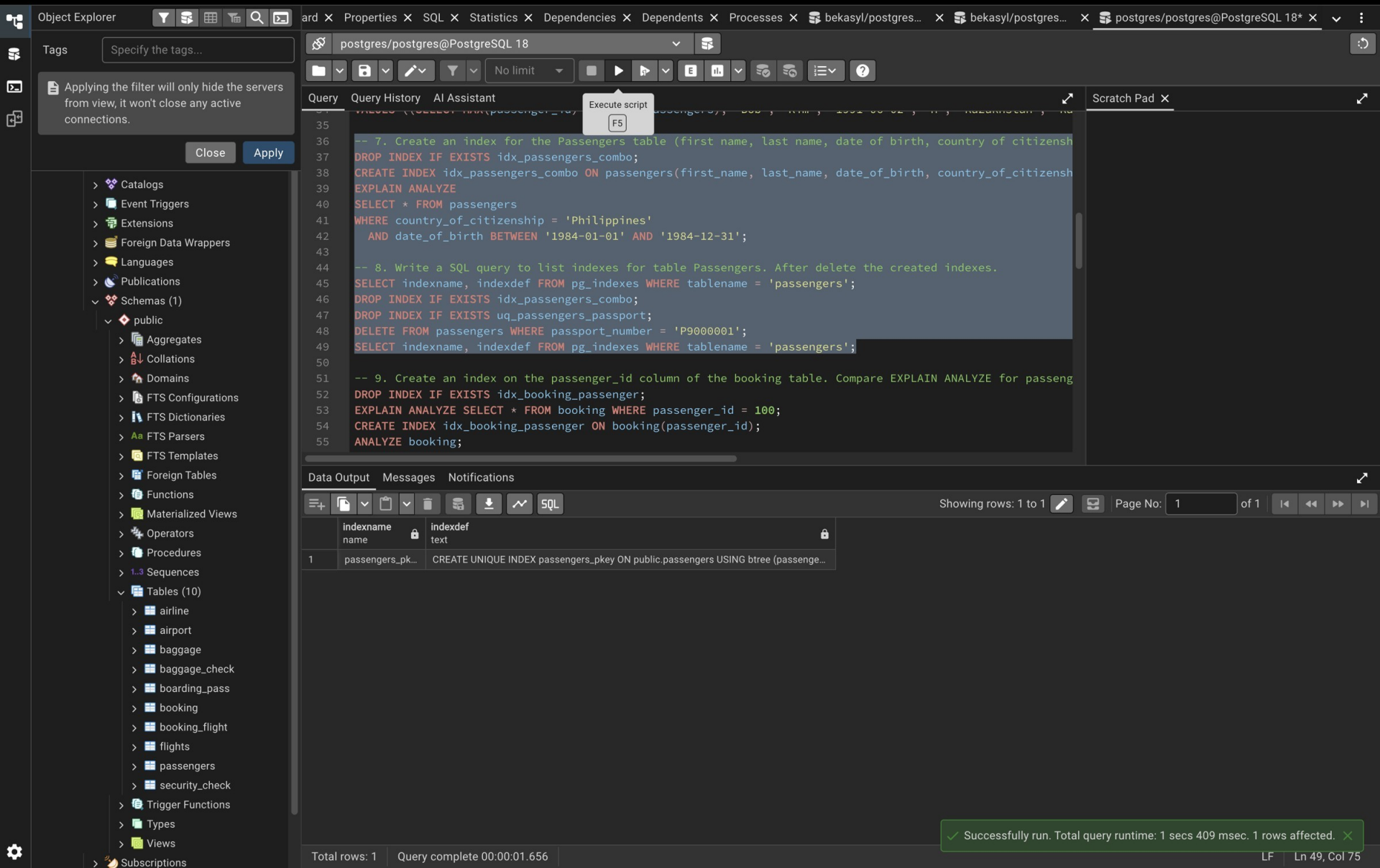
Tasks 1, 2, 3


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Applying the filter will only hide the servers from view, it won't close any active connections.
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bekasyl/postgres...
bekasyl/postgres...
postgres/postgres@PostgreSQL 18*
postgres/postgres@PostgreSQL 18
No limit
Query
Query History
AI Assistant
Execute script
F5
Scratch Pad
1 ANALYZE;
2
3 -- 1. Create an index on the actual_departure column in the flights table.
4 CREATE INDEX idx_flights_actual_departure ON flights(act_departure_time);
5
6 -- 2. Create a unique index to ensure flight_no and scheduled_departure combinations are unique.
7 CREATE UNIQUE INDEX uq_flights_no_sched ON flights(airline_id, departing_airport_id, arriving_airport_id, s
8
9 -- 3. Create a composite index on the departure_airport_id and arrival_airport_id columns.
10 CREATE INDEX idx_flights_dep_arr ON flights(departing_airport_id, arriving_airport_id);
11
12 -- 4. Evaluate the difference in query performance with and without indexes. Measure performance difference
13 DROP INDEX idx_flights_dep_arr;
14 EXPLAIN ANALYZE SELECT * FROM flights WHERE departing_airport_id = 10 AND arriving_airport_id = 20;
15 CREATE INDEX idx_flights_dep_arr ON flights(departing_airport_id, arriving_airport_id);
16 ANALYZE flights;
17 EXPLAIN ANALYZE SELECT * FROM flights WHERE departing_airport_id = 10 AND arriving_airport_id = 20;
18
19 -- 5. Use EXPLAIN ANALYZE to check index usage in a query filtering by departure_airport and arrival_airpor
20 EXPLAIN ANALYZE SELECT * FROM flights WHERE departing_airport_id = 10 AND arriving_airport_id = 20;
21
Data Output
Messages
Notifications
CREATE INDEX
Query returned successfully in 8 secs 376 msec.
Total rows:
Query complete 00:00:08.376
LF
Ln 10, Col 88

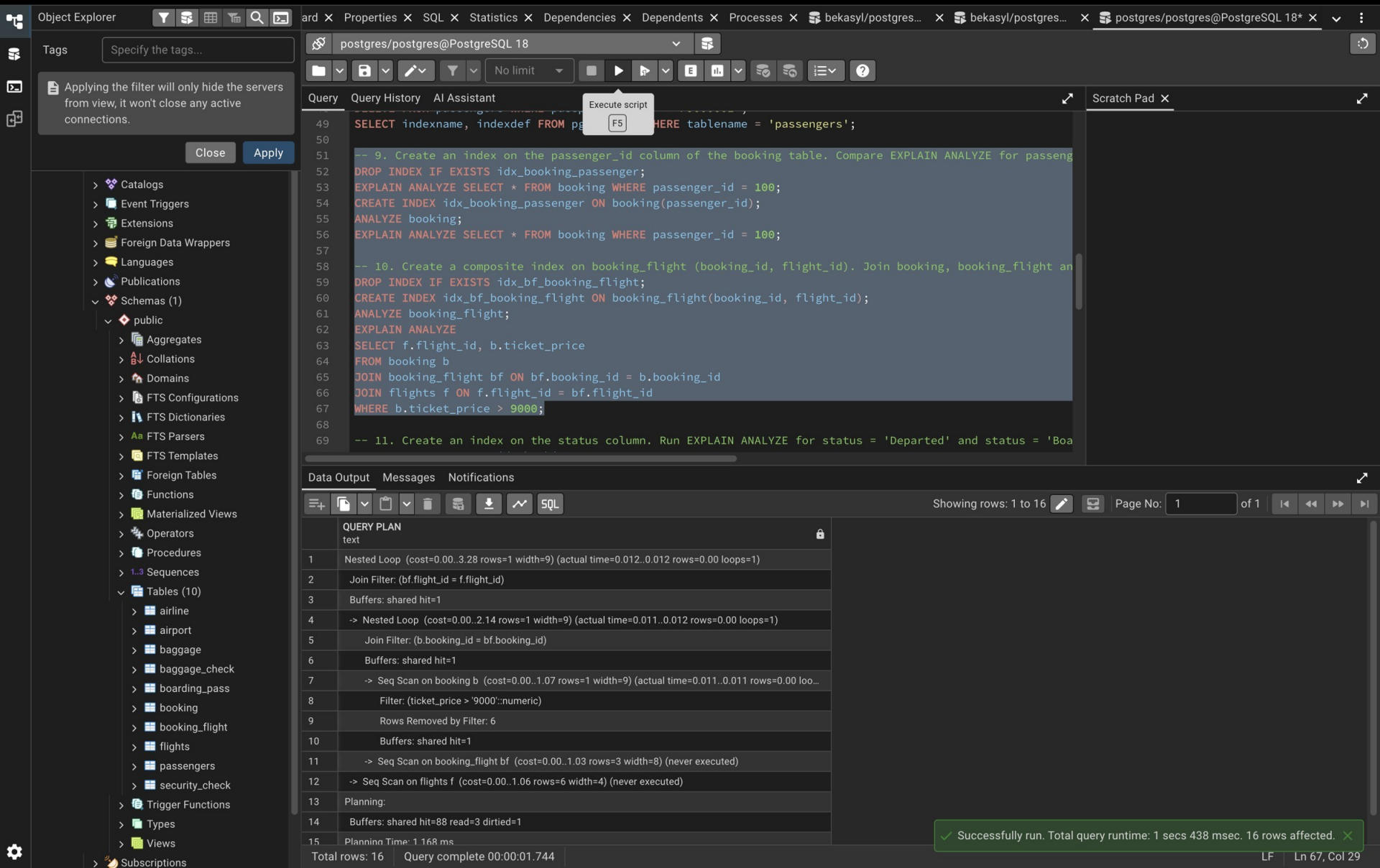
Tasks 4, 5, 6


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Processes
postgres/postgres@PostgreSQL 18*
postgres/postgres@PostgreSQL 18
No limit
Query
Query History
AI Assistant
Scratch Pad
10 CREATE INDEX idx_flights_dep_arr ON flights(departing_airport_id, arriving_airport_id);
11
12 -- 4. Evaluate the difference in query performance with and without indexes. Measure performance difference
13 DROP INDEX idx_flights_dep_arr;
14 EXPLAIN ANALYZE SELECT * FROM flights WHERE departing_airport_id = 10 AND arriving_airport_id = 20;
15 CREATE INDEX idx_flights_dep_arr ON flights(departing_airport_id, arriving_airport_id);
16 ANALYZE flights;
17 EXPLAIN ANALYZE SELECT * FROM flights WHERE departing_airport_id = 10 AND arriving_airport_id = 20;
18
19 -- 5. Use EXPLAIN ANALYZE to check index usage in a query filtering by departure_airport and arrival_airpor
20 EXPLAIN ANALYZE SELECT * FROM flights WHERE departing_airport_id = 10 AND arriving_airport_id = 20;
21
22 -- 6. Create a unique index for the passport_number of the Passengers table. Check if the index was created
23 CREATE UNIQUE INDEX uq_passengers_passport ON passengers(passport_number);
24 SELECT indexname, indexdef FROM pg_indexes WHERE tablename = 'passengers';
25 INSERT INTO passengers (passenger_id, first_name, last_name, date_of_birth, gender, country_of_citizenship,
26 VALUES ((SELECT MAX(passenger_id) + 1 FROM passengers), 'Ann', 'Lee', '1990-05-01', 'F', 'Kazakhstan', 'Kaz
27 INSERT INTO passengers (passenger_id, first_name, last_name, date_of_birth, gender, country_of_citizenship,
28 VALUES ((SELECT MAX(passenger_id) + 1 FROM passengers), 'Bob', 'Kim', '1991-06-02', 'M', 'Kazakhstan', 'Kaz
29
30 -- 7. Create an index for the Passengers table (first name, last name, date of birth, country of citizenshi
31 CREATE INDEX idx_passengers_combo ON passengers(first_name, last_name, date_of_birth, country_of_citizenshi
Data Output
Messages
Notifications
ERROR: duplicate key value violates unique constraint "uq_passengers_passport"
Key (passport_number)=(P9000001) already exists.
SQL state: 23505
Detail: Key (passport_number)=(P9000001) already exists.
Total rows:
Query complete 00:00:02.184
LF
Ln 29, Col 1

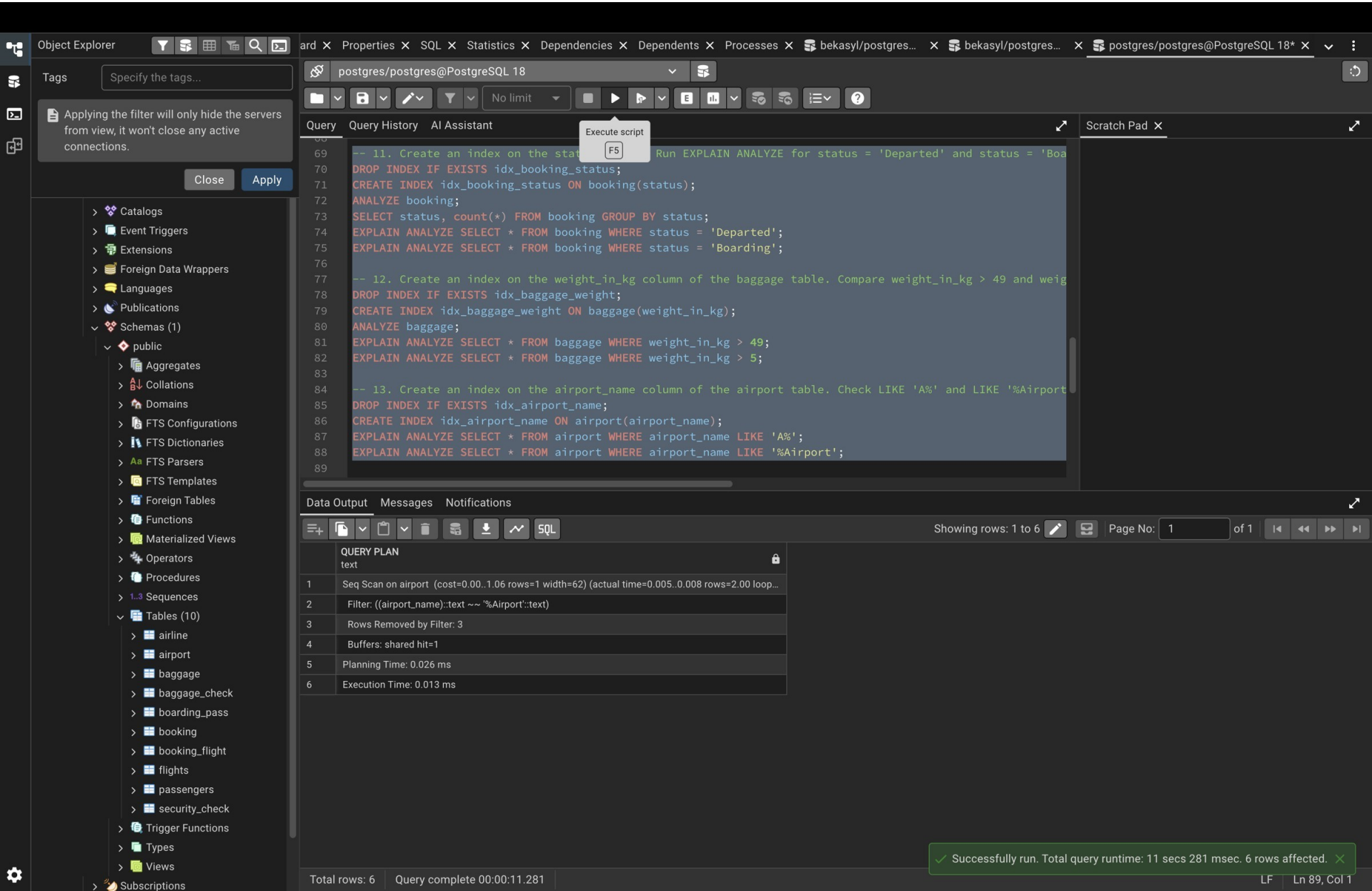
Tasks 7, 8


Object Explorer
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bekasyl/postgres...
bekasyl/postgres...
postgres/postgres@PostgreSQL 18*
postgres/postgres@PostgreSQL 18
No limit
Query
Query History
AI Assistant
Execute script
F5
Scratch Pad
-- 7. Create an index for the Passengers table (first name, last name, date of birth, country of citizensh
DROP INDEX IF EXISTS idx_passengers_combo;
CREATE INDEX idx_passengers_combo ON passengers(first_name, last_name, date_of_birth, country_of_citizensh
EXPLAIN ANALYZE
SELECT * FROM passengers
WHERE country_of_citizenship = 'Philippines'
AND date_of_birth BETWEEN '1984-01-01' AND '1984-12-31';
-- 8. Write a SQL query to list indexes for table Passengers. After delete the created indexes.
SELECT indexname, indexdef FROM pg_indexes WHERE tablename = 'passengers';
DROP INDEX IF EXISTS idx_passengers_combo;
DROP INDEX IF EXISTS uq_passengers_passport;
DELETE FROM passengers WHERE passport_number = 'P9000001';
SELECT indexname, indexdef FROM pg_indexes WHERE tablename = 'passengers';
-- 9. Create an index on the passenger_id column of the booking table. Compare EXPLAIN ANALYZE for passeng
DROP INDEX IF EXISTS idx_booking_passenger;
EXPLAIN ANALYZE SELECT * FROM booking WHERE passenger_id = 100;
CREATE INDEX idx_booking_passenger ON booking(passenger_id);
ANALYZE booking;
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 1
Page No: 1 of 1
indexname name
indexdef text
1
passengers_pk...
CREATE UNIQUE INDEX passengers_pkey ON public.passengers USING btree (passenge...
Successfully run. Total query runtime: 1 secs 409 msec. 1 rows affected.
Total rows: 1
Query complete 00:00:01.656
LF
Ln 49, Col 75

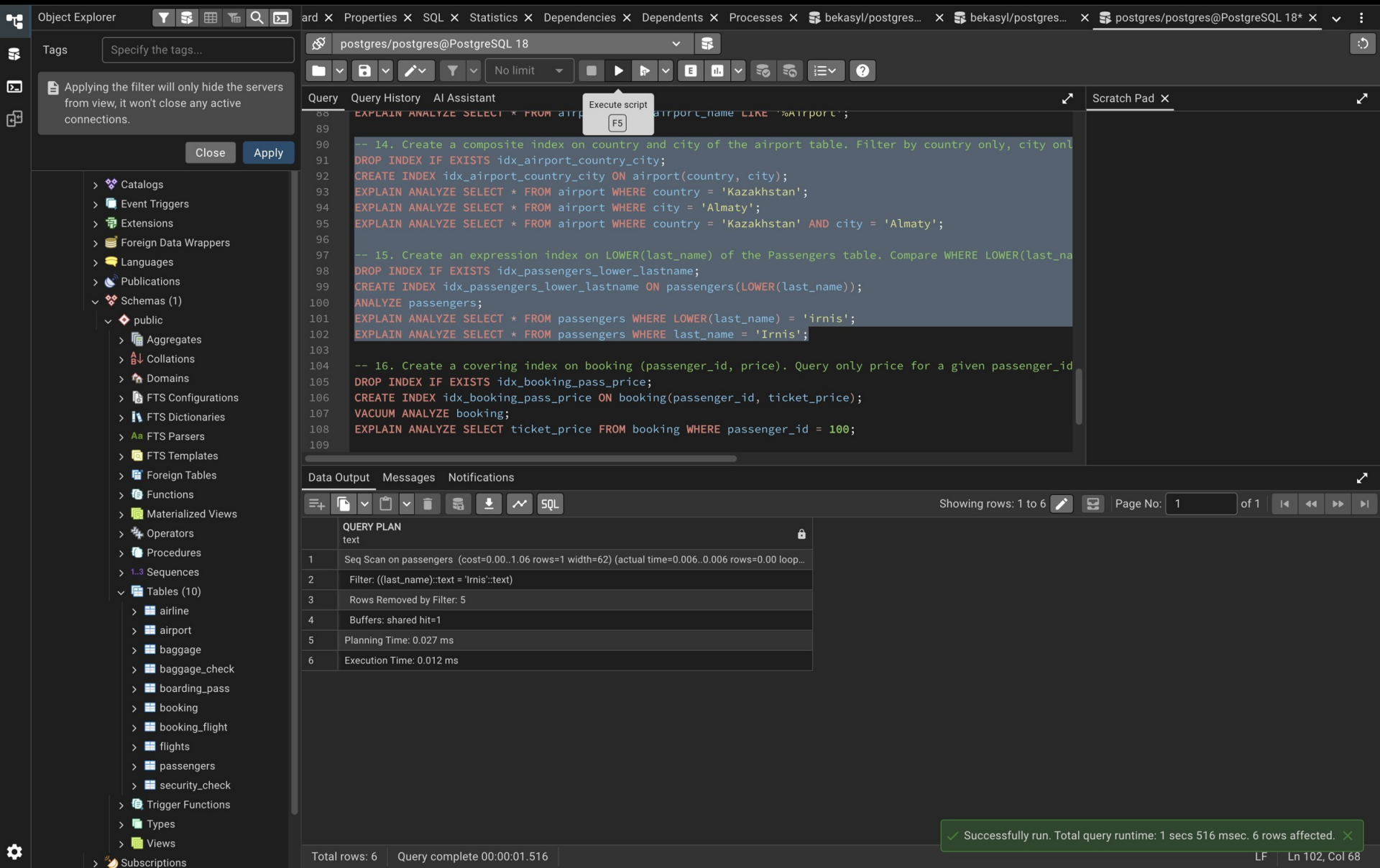
Tasks 9, 10


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bekasyl/postgres...
bekasyl/postgres...
postgres/postgres@PostgreSQL 18*
postgres/postgres@PostgreSQL 18
No limit
Query
Query History
AI Assistant
Execute script
F5
Scratch Pad
49 SELECT indexname, indexdef FROM pg HERE tablename = 'passengers';
50
51 -- 9. Create an index on the passenger_id column of the booking table. Compare EXPLAIN ANALYZE for passeng
52 DROP INDEX IF EXISTS idx_booking_passenger;
53 EXPLAIN ANALYZE SELECT * FROM booking WHERE passenger_id = 100;
54 CREATE INDEX idx_booking_passenger ON booking(passenger_id);
55 ANALYZE booking;
56 EXPLAIN ANALYZE SELECT * FROM booking WHERE passenger_id = 100;
57
58 -- 10. Create a composite index on booking_flight (booking_id, flight_id). Join booking, booking_flight an
59 DROP INDEX IF EXISTS idx_bf_booking_flight;
60 CREATE INDEX idx_bf_booking_flight ON booking_flight(booking_id, flight_id);
61 ANALYZE booking_flight;
62 EXPLAIN ANALYZE
63 SELECT f.flight_id, b.ticket_price
64 FROM booking b
65 JOIN booking_flight bf ON bf.booking_id = b.booking_id
66 JOIN flights f ON f.flight_id = bf.flight_id
67 WHERE b.ticket_price > 9000;
68
69 -- 11. Create an index on the status column. Run EXPLAIN ANALYZE for status = 'Departed' and status = 'Boa
Data Output
Messages
Notifications
SQL
Showing rows: 1 to 16
Page No: 1 of 1
QUERY PLAN
text
1 Nested Loop (cost=0.00..3.28 rows=1 width=9) (actual time=0.012..0.012 rows=0.00 loops=1)
2 Join Filter: (bf.flight_id = f.flight_id)
3 Buffers: shared hit=1
4 -> Nested Loop (cost=0.00..2.14 rows=1 width=9) (actual time=0.011..0.012 rows=0.00 loops=1)
5 Join Filter: (b.booking_id = bf.booking_id)
6 Buffers: shared hit=1
7 -> Seq Scan on booking b (cost=0.00..1.07 rows=1 width=9) (actual time=0.011..0.011 rows=0.00 loo...
8 Filter: (ticket_price > '9000'::numeric)
9 Rows Removed by Filter: 6
10 Buffers: shared hit=1
11 -> Seq Scan on booking_flight bf (cost=0.00..1.03 rows=3 width=8) (never executed)
12 -> Seq Scan on flights f (cost=0.00..1.06 rows=6 width=4) (never executed)
13 Planning:
14 Buffers: shared hit=88 read=3 dirtied=1
15 Planning Time: 1.168 ms
Successfully run. Total query runtime: 1 secs 438 msec. 16 rows affected.
Total rows: 16
Query complete 00:00:01.744
LF
Ln 67, Col 29

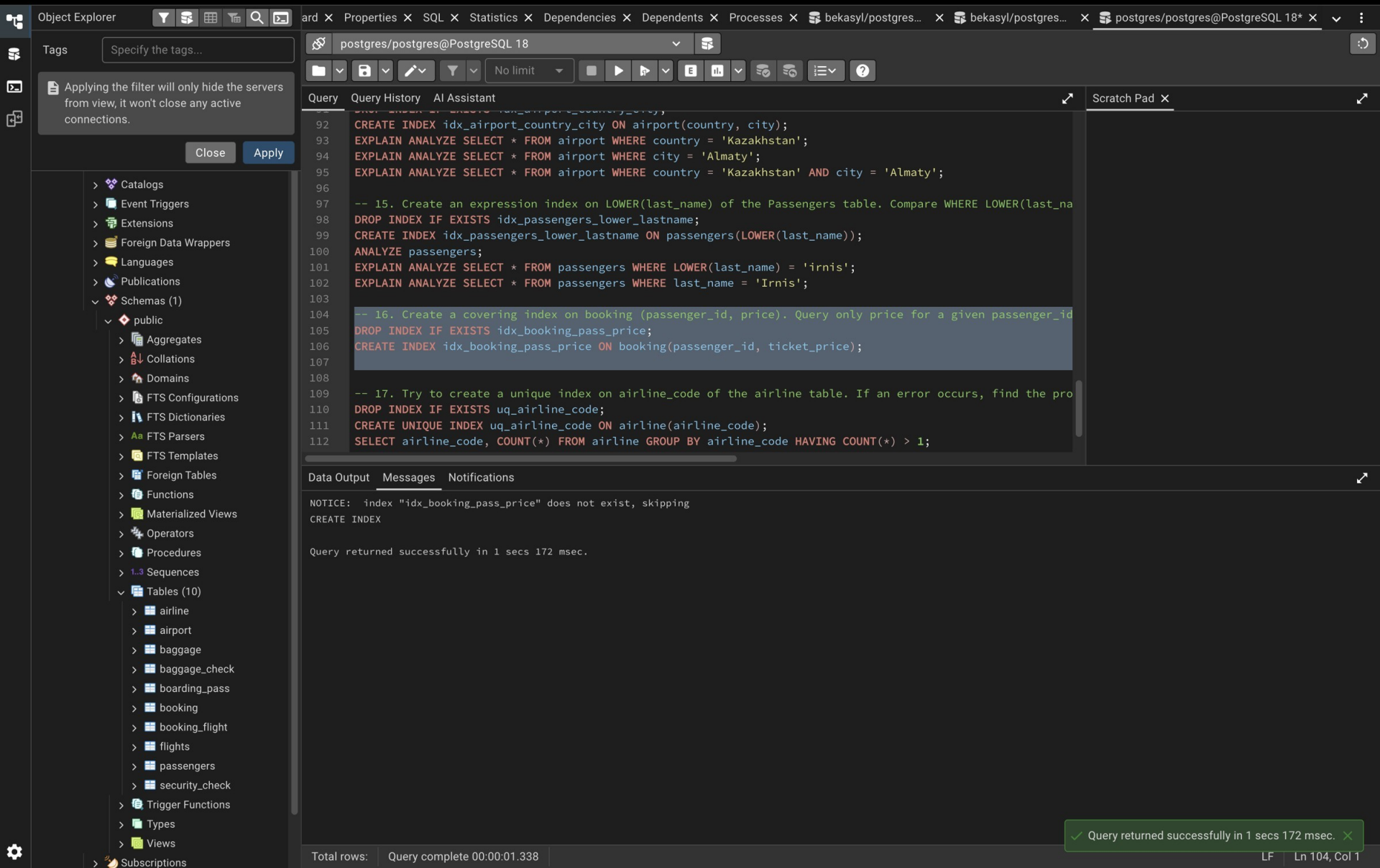
Tasks 11, 12, 13



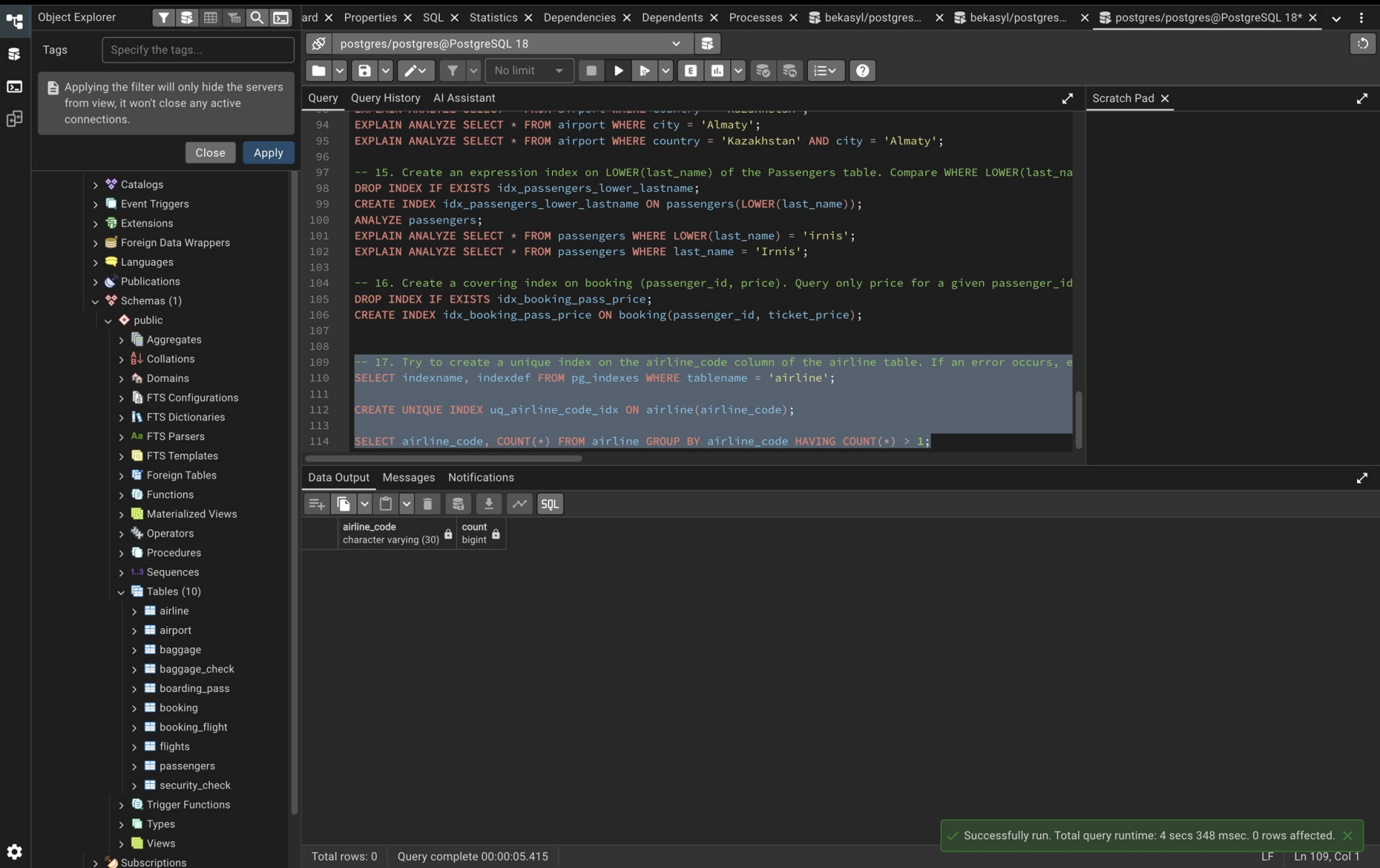
Tasks 14, 15, 16



Task 16


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bekasyl/postgres...
postgres/postgres@PostgreSQL 18*
postgres/postgres@PostgreSQL 18
No limit
Query
Query History
AI Assistant
Scratch Pad
92 CREATE INDEX idx_airport_country_city ON airport(country, city);
93 EXPLAIN ANALYZE SELECT * FROM airport WHERE country = 'Kazakhstan';
94 EXPLAIN ANALYZE SELECT * FROM airport WHERE city = 'Almaty';
95 EXPLAIN ANALYZE SELECT * FROM airport WHERE country = 'Kazakhstan' AND city = 'Almaty';
96
97 -- 15. Create an expression index on LOWER(last_name) of the Passengers table. Compare WHERE LOWER(last_na
98 DROP INDEX IF EXISTS idx_passengers_lower_lastname;
99 CREATE INDEX idx_passengers_lower_lastname ON passengers(LOWER(last_name));
100 ANALYZE passengers;
101 EXPLAIN ANALYZE SELECT * FROM passengers WHERE LOWER(last_name) = 'irnis';
102 EXPLAIN ANALYZE SELECT * FROM passengers WHERE last_name = 'Irnis';
103
104 -- 16. Create a covering index on booking (passenger_id, price). Query only price for a given passenger_id
105 DROP INDEX IF EXISTS idx_booking_pass_price;
106 CREATE INDEX idx_booking_pass_price ON booking(passenger_id, ticket_price);
107
108
109 -- 17. Try to create a unique index on airline_code of the airline table. If an error occurs, find the pro
110 DROP INDEX IF EXISTS uq_airline_code;
111 CREATE UNIQUE INDEX uq_airline_code ON airline(airline_code);
112 SELECT airline_code, COUNT(*) FROM airline GROUP BY airline_code HAVING COUNT(*) > 1;
Data Output
Messages
Notifications
NOTICE: index "idx_booking_pass_price" does not exist, skipping
CREATE INDEX
Query returned successfully in 1 secs 172 msec.
Query returned successfully in 1 secs 172 msec.
Total rows:
Query complete 00:00:01.338
LF
Ln 104, Col 1

Task 17


Object Explorer
Tags
Specify the tags...
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bekasyl/postgres...
bekasyl/postgres...
postgres/postgres@PostgreSQL 18*
postgres/postgres@PostgreSQL 18
No limit
Query
Query History
AI Assistant
Scratch Pad
94 EXPLAIN ANALYZE SELECT * FROM airport WHERE city = 'Almaty';
95 EXPLAIN ANALYZE SELECT * FROM airport WHERE country = 'Kazakhstan' AND city = 'Almaty';
96
97 -- 15. Create an expression index on LOWER(last_name) of the Passengers table. Compare WHERE LOWER(last_na
98 DROP INDEX IF EXISTS idx_passengers_lower_lastname;
99 CREATE INDEX idx_passengers_lower_lastname ON passengers(LOWER(last_name));
100 ANALYZE passengers;
101 EXPLAIN ANALYZE SELECT * FROM passengers WHERE LOWER(last_name) = 'irnis';
102 EXPLAIN ANALYZE SELECT * FROM passengers WHERE last_name = 'Irnis';
103
104 -- 16. Create a covering index on booking (passenger_id, price). Query only price for a given passenger_id
105 DROP INDEX IF EXISTS idx_booking_pass_price;
106 CREATE INDEX idx_booking_pass_price ON booking(passenger_id, ticket_price);
107
108
109 -- 17. Try to create a unique index on the airline_code column of the airline table. If an error occurs, e
110 SELECT indexname, indexdef FROM pg_indexes WHERE tablename = 'airline';
111
112 CREATE UNIQUE INDEX uq_airline_code_idx ON airline(airline_code);
113
114 SELECT airline_code, COUNT(*) FROM airline GROUP BY airline_code HAVING COUNT(*) > 1;
Data Output
Messages
Notifications
SQL
airline_code
character varying (30)
count
bigint
Successfully run. Total query runtime: 4 secs 348 msec. 0 rows affected.
Total rows: 0
Query complete 00:00:05.415
LF
Ln 109, Col 1